Process Characterization Report

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCR-006

Process Development / Manufacturing Science & Technology (MSAT)

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Reviewer	Analytical Development	—	—	—
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Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; RSM, response-surface methodology; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; TCID50, fifty-percent tissue-culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Low-pH viral inactivation is a hold step, and it is the operation on which the enveloped-virus clearance claim for A-Mab drug substance principally rests. The Protein A eluate is titrated into an acidic hold, held for a defined time under temperature control, and neutralized before it is loaded onto cation exchange. This report presents the characterization executed under PCP-006 on a qualified small-scale spiking model, with the scope set by the risk assessment in RA-001. The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development principles of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), and the clearance claim is framed as a modular claim under ICH Q5A(R2) (International Council for Harmonisation 2023a) in the manner of the case study on which the process is modelled (CMC Biotech Working Group 2009).

Four parameters were characterized. Inactivation pH, hold time and temperature were studied together in a multivariate design of 11 screening runs and 18 response-surface runs. A-Mab concentration was assessed one factor at a time. The screening design identified all three multivariate parameters as active on the XMuLV log reduction factor and found no significant interaction among them. The response-surface model is the predictive model, and at the set-point it predicts a step log reduction factor of 6.87 $\log_{10}$ against a required step contribution of 4.93 $\log_{10}$.

Inactivation pH is classified as a critical process parameter, and it is the only parameter in the drug substance process to carry that classification. Its characterized range fails in both directions. Toward the top, clearance falls: the model predicts a decline of 1.33 $\log_{10}$ between the set-point and the top of the range, and at the corner formed by the highest pH, the shortest hold and the lowest temperature the prediction falls 0.02 $\log_{10}$ short of the required contribution. Toward the bottom, the range stops where platform data place the onset of acid-induced aggregation, and no run was executed below that pH. Hold time and temperature are classified as well-controlled critical process parameters and A-Mab concentration as a general process parameter.

The step raises aggregate instead of reducing it, and the increase is governed by hold time. Predicted pool aggregate rises from 1.79 % to 2.27 % across the characterized hold-time range at set-point pH and temperature, against a drug substance limit of 5 %. Acidic variants rise by 0.4 percentage points over the same interval. Neither attribute approaches its limit anywhere in the characterized ranges. Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches operated within their normal operating ranges. Cumulative XMuLV clearance is the tightest of the three, with a one-sided Cpk of 1.53 and a mean of 19.51 $\log_{10}$ against a requirement of 16.7 $\log_{10}$.

Two deviations were recorded, investigated and closed. Both were retained, and neither altered a fitted effect, the operating region or a classification. Two claims are explicitly not made here. No clearance of non-enveloped virus is claimed, and the credited MVM contribution of this step is 0.00 $\log_{10}$, so parvovirus clearance rests on anion exchange (PCR-008) and virus filtration (PCR-009). No host cell protein clearance is claimed either, because HCP was not measured across this step. The step is understood over the ranges studied, its operating region is defined in Section 6, and the outcome rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch cell culture and purified by a platform train that has been used for three earlier antibodies (Section 2.1). The train runs from harvest and clarification through Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. Table 1.1 lists the steps in order with the role each plays in the control strategy.

Table 1.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Low-pH viral inactivation is Step 6 and the first of the three operations that carry a viral clearance claim (Steps 6, 8 and 9). The Protein A eluate arrives acidic from the elution buffer, and the step titrates it to the target inactivation pH, holds it for a defined time under temperature control, and then neutralizes the pool for the cation exchange load. The mechanism is chemical and not physical, in that protonation at low pH disrupts the lipid envelope and the envelope glycoproteins of an enveloped virus, and the resulting loss of infectivity is not reversed on neutralization. A non-enveloped virus has no envelope to disrupt, which is why the step is credited for XMuLV and not for MVM.

The step forms no product quality attribute, in that no glycan, charge or size attribute is created here and the pool that leaves the step contains the same molecule that entered it. What the step does set is the cumulative enveloped-virus clearance of the drug substance, an attribute of very high criticality (Table 2.1), and it acts on two attributes formed upstream at the bioreactor (PCR-003). The acid hold raises soluble aggregate and acidic charge variants slightly, and both were measured as responses of this study alongside the log reduction factor.

1.2 Objectives

The study had five objectives.

(i) Quantify the relationship between the operating parameters of the step and the XMuLV log reduction factor, pool aggregate and acidic charge variants, over ranges wider than those intended for routine operation.

(ii) Define a multivariate operating region within which the step delivers its required contribution to enveloped-virus clearance.

(iii) Establish proven acceptable ranges for each parameter against each quality attribute it governs.

(iv) Classify every parameter of the step under SOP-4001 (CPP, WC-CPP, KPP or GPP).

(v) Justify the normal operating ranges and the elements of the control strategy that this step contributes for Stage 2 operation at the receiving site.

1.3 Regulatory and scientific basis

This report is the Stage 1 process design record for the step in the sense of the lifecycle guidance on process validation (U.S. Food and Drug Administration 2011). Criticality is treated as a continuum and not as a binary classification, following the quality risk management principles of ICH Q9 (International Council for Harmonisation 2023b), and the enhanced development approach of ICH Q8 and ICH Q11 supplies the design space and control strategy framework (International Council for Harmonisation 2009, 2012). Viral clearance is evaluated under ICH Q5A(R2) (International Council for Harmonisation 2023a) as a modular claim, in which each clearance step is spiked and studied on its own and the step contributions are summed. The process description, the quality attribute framework and the risk methodology follow the published A-Mab case study (CMC Biotech Working Group 2009).

Two features of the viral-safety framework shape what this report is able to claim. Clearance studies are performed at small scale with deliberately spiked virus, because live virus may not be introduced into a manufacturing facility, so the claim rests on the qualification of the scale-down model and never on a manufacturing measurement. Clearance is also credited only where the mechanism is understood and the step is operated inside a range that has been studied, which is why the operating region in Section 6 is expressed in the same three parameters as the clearance claim itself.

1.4 Related documents

Table 1.2 lists the documents that scope this study and the documents into which it feeds.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-006	Process Characterization Plan (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The low-pH hold is the most stable operation in the platform. It has been used essentially unchanged for three earlier humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) and through A-Mab clinical and engineering manufacture at the sending site. The buffer system, the hold configuration and the parameter set have not changed across those products. Platform experience therefore establishes the mechanism and the direction of every effect this study expected to see, and the purpose of the characterization was to bound those effects for A-Mab and not to discover them.

The mechanistic expectations were recorded in PCP-006 before execution. Inactivation of an enveloped virus by acid is a kinetic process, so the log reduction factor was expected to rise with hold time and to fall as pH rises toward neutrality. Rate constants for chemical denaturation rise with temperature, so a warmer hold was expected to inactivate faster. Unfolding of an IgG1 at low pH exposes hydrophobic surface and promotes association, so aggregate was expected to rise with hold time and with temperature, and to rise sharply once pH falls below the stability edge of the molecule. Asparagine deamidation is only weakly acid-catalysed at these pH values, so acidic variants were expected to drift upward slowly across the hold and to be insensitive to pH over a range this narrow.

Two of these expectations set the edges of the characterization range directly. Univariate platform work indicated that A-Mab begins to aggregate on prolonged exposure below pH 3.2, so that value was taken as the lowest pH studied and no run was executed below it. At the other edge, platform experience did not support complete inactivation within a normal hold at pH values approaching neutrality, and the upper edge was set at pH 4.0 in order to place the expected edge of failure inside the studied range instead of outside it.

2.2 Quality attributes in scope

Three quality attributes were measured across the step. Table 2.1 gives their acceptance criteria, their criticality and their Tool #1 scores.

Table 2.1: Quality attributes measured across the low-pH viral inactivation step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Enveloped-virus clearance is the attribute the step sets, and it is the only attribute in the table whose acceptance criterion is cumulative across the process, and it is not a property of the pool leaving this step. It is assigned very high criticality. Its Tool #1 score of 20 is lower than the score carried by aggregate, because Tool #1 combines the impact of an attribute with the uncertainty attached to that impact, and the consequence of insufficient viral clearance is understood with very little uncertainty. The Tool #2 severity of 9 is the highest assigned to any attribute of the drug substance.

Aggregate and acidic variants are formed at the bioreactor and are carried into this step by the Protein A pool. Neither is set here. Both were measured because the acid hold is a recognized source of both, and because a hold long enough to guarantee clearance must be shown not to damage the product. Aggregate is of high criticality and acidic variants of very low criticality, which is why the discussion of the two in Section 5 is asymmetric.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the step twice, once for its potential impact on quality attributes and once for its potential to interact with other parameters, and assigned a study type from the combined score. Three parameters ranked high enough to require multivariate study and one was assigned to univariate assessment. Table 4.1 gives the outcome together with the ranges used, and the reasoning behind each range is in Section 4.1.

The ranges were widened well beyond the intended operating ranges, so that the study would locate the edges of acceptable performance instead of confirming the centre. The characterized ranges are between 2.0 and 4.0 times as wide as the corresponding normal operating ranges. That choice is what makes the parameter classification in Section 9 an argument from data, because a range that does not reach an edge of failure cannot distinguish a critical parameter from a well-controlled one.

3 Materials and methods

3.1 Scale-down model and its qualification

Viral clearance cannot be measured on a manufacturing batch, so the entire claim rests on a small-scale model and on the evidence that the model represents the commercial step, which for this study is a scaled hold vessel operated under SOP-2009 and qualified under SOP-1001. Spiking, sampling and titration were performed under the same procedures (SOP-2009) for every run of both designs.

The model is a chemical replica of the step and not a geometric one. Titration uses the same acid and the same buffer system at the same molar ratio, the target pH is reached over a comparable ramp, and the hold is thermostatted to its set-point throughout. Inactivation pH, hold time and temperature are scale-independent quantities and were operated at the values that commercial manufacture will use. Mixing time and titration ramp are scale-dependent, and the small-scale conditions were set to match the ramp achieved at manufacturing scale, not an impeller speed.

Qualification compared the model with the at-scale step on input and output attributes measured by the same validated methods, including pool aggregate (AMV-3011), acidic variants (AMV-3013) and the pH achieved at the end of titration. No difference was found that would alter the effects fitted in this report. The qualification record is held under SOP-1001, and this report cites its conclusion without restating its data.

One boundary of the scale-down warrant is stated here because it limits Section 6. The model reproduces the chemistry of the hold and the clearance that the chemistry delivers. It does not reproduce the geometry of the commercial hold vessel, so it carries no information about mixing homogeneity during titration at scale. That question is addressed by the equipment qualification and comparability work described in PTP-001.

3.2 Operation

The step receives the Protein A eluate directly and without adjustment of protein concentration. Acid is added under pH control until the target inactivation pH (Table 4.1) is reached, the pH is confirmed by an independent measurement, and only then does the hold clock start. The hold proceeds without further addition. At the end of the hold the pool is titrated to the cation exchange load pH and filtered, and neutralization is treated as the end of the inactivation for the purpose of the log reduction calculation.

A-Mab concentration in the hold is set by the Protein A elution and is not adjusted at this step. It was therefore treated as an input attribute spanning the range that Protein A can deliver, which is why it appears in Table 4.1 with a normal operating range equal to its characterized range.

Materials entering the step are controlled to platform specifications and were not characterized. Acid and neutralization buffers were prepared and released under SOP-2103 before use, and their identity, concentration and expiry are quality controlled, and none of them was studied as a parameter.

3.3 Analytical methods

Table 3.1 lists the controlled procedures and validated methods used in the study.

Table 3.1: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

XMuLV infectivity was measured as a TCID50 titre under AMV-3017, and the log reduction factor is the difference between the titre of the spiked load and the titre of the neutralized hold pool. Aggregate was measured by SEC-HPLC under AMV-3011 and charge variants by icIEF under AMV-3013. Each assay was run with the cytotoxicity and interference controls that ICH Q5A(R2) requires for a clearance study (International Council for Harmonisation 2023a).

The infectivity assay is the least precise of the three and it is the method that carries the claim, so two of its properties are stated here and used in Section 5. Its quantitation limit is 1.5 $\log_{10}$ TCID50/mL, and 28 % of the variance observed in the reported log reduction factor is attributable to the method itself. Method performance is summarized in Appendix D and reported in full in the validation record for AMV-3017.

3.4 Sampling plan

Each run was sampled twice. The load sample was taken after the virus spike had been added and after the target pH had been reached, and it establishes the input titre. The pool sample was taken at the end of the hold, neutralized immediately to stop further inactivation, and assayed for infectivity, aggregate and charge variants. Sampling volumes were small enough that the hold volume was not materially changed.

Replication was placed at the centre point of each design. The screening design carried 3 centre-point runs and the response-surface design carried 4, giving 2 and 3 degrees of freedom for pure error respectively. Those replicates are the only estimate of reproducibility between runs in the study, and every lack-of-fit test in Section 5 is made against them.

3.5 Statistical methods

Factors were coded to the edges of their characterized ranges, with the coded values −1 and +1 at the range limits and 0 at the range midpoint. The midpoint is not the process set-point for any of the three multivariate factors (Table 4.1), so every prediction quoted at the set-point in this report was computed from the fitted model at the set-point values and not read off the design centre. Analyses were performed under SOP-1002.

The screening analysis fitted all main effects and all two-factor interactions by ordinary least squares. An effect is reported as twice the coded coefficient, so it is the change in the response across the full characterized range of that factor. The response-surface analysis fitted the full quadratic model in the same three factors. Significance is assessed at the 5 % level throughout, and coefficients are reported with their standard errors.

Model adequacy was judged on the coefficient of determination and its adjusted form, on the predicted coefficient of determination computed from PRESS, on the overall F test, and on an analysis of variance that partitions the residual sum of squares into lack of fit and pure error. A model is treated as adequate when its lack-of-fit term is not significant against the centre-point pure error. A response for which no factor is active is reported as robust over the ranges studied and is not modelled further.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with every process parameter drawn within its normal operating range and propagated through the whole process model. Capability is reported as a one-sided Cpk, because each attribute this step influences has a single binding limit. Proven acceptable ranges were computed in the two forms described in Section 7.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the four parameters of the step with their set-points, normal operating ranges, characterized ranges and final classifications. Table 4.2 gives the letter codes used for the design factors in the effect and coefficient tables that follow.

Table 4.1: Parameters of the low-pH viral inactivation step, with set-points, normal operating ranges, characterization ranges and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Inactivation pH	pH	3.5	3.4–3.6	3.2–4	CPP	multivariate
Hold time	min	90	60–120	60–180	WC-CPP	multivariate
Temperature	°C	21	19–23	15–25	WC-CPP	multivariate
A-Mab concentration	g/L	20	5–35	5–35	GPP	univariate

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

The characterized ranges are the knowledge space of the step. The normal operating ranges lie inside them, and the operating region defined in Section 6 lies between the two. Each range was set for a stated reason, and no uniform multiple of the operating range was applied.

The pH range is not centred on the set-point, and it extends 0.3 pH units below the set-point and 0.5 units above it, because the edge of failure that the study needed to locate is the upper one. The lower edge was placed at the pH below which platform data indicate aggregation, and no run went below it.

The hold-time range extends to 180 minutes, which is 2 times the set-point hold. A hold can be extended in manufacture but cannot be shortened once it has run, so the practical risk lies at the long end and the range was widened in that direction. Its lower edge coincides with the lower edge of the normal operating range, because a hold shorter than that is a batch deviation and not an operating condition. The temperature range spans the ambient control capability of the receiving suite (Grafton, WI), which is 10 °C wide against a normal operating range of 4 °C.

4.2 Screening design

The screening study was a two-level full factorial in the three multivariate parameters with 3 centre points, 11 runs in total. A full factorial in three factors estimates all three main effects and all three two-factor interactions without aliasing, so no resolution compromise applies and no effect in Table 5.3 is confounded with another. Runs were randomized and executed under SOP-1002 on the qualified scale-down model. The coded design matrix and the measured responses are in Appendix A.

The screening model leaves 4 residual degrees of freedom, and it contains no quadratic term. It identifies which factors are active and estimates their magnitude approximately, and it cannot detect curvature. It is not used as a predictive model anywhere in this report.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same three factors, with 8 factorial runs, 6 axial runs at the face centres and 4 centre points, 18 runs in total. Placing the axial points on the faces of the cube, instead of outside it, keeps every run inside the characterized ranges, which matters here because the ranges were chosen to reach an edge of failure and an extrapolated axial point would have crossed it. The design supports the full quadratic model in three factors. That model is the predictive model of this report and the basis of the operating region in Section 6, the proven acceptable ranges in Section 7 and the capability assessment in Section 8. The coded matrix and responses are in Appendix B.

All three factors were carried into the response-surface design, because with only three multivariate parameters there was no subset to select and curvature in the pH response was the specific question the design was built to answer.

4.4 Univariate assessment

A-Mab concentration was assessed one factor at a time and was not carried into the multivariate design. Acid-mediated inactivation of an enveloped virus is a reaction between protons and the viral envelope, and the antibody is not a reactant in it, so no mechanism links antibody concentration to the inactivation rate. RA-001 ranked the parameter on that basis and assigned it to univariate assessment.

Table 4.1 records the outcome. The normal operating range for A-Mab concentration is its entire characterized range, and that equality is the record that no restriction on the parameter was found to be necessary. The range covers the concentrations that the Protein A step can deliver, so this step imposes no additional control on the eluate it receives.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility differs sharply between the three responses, and it sets how far each result can be pushed. Table 5.1 gives the mean, standard deviation and coefficient of variation of the 4 centre-point runs of the response-surface design.

Table 5.1: Centre-point reproducibility of the response-surface design (pure-error basis for the lack-of-fit tests).

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	2.006	0.01694	0.8446
XMuLV LRF (log10)	4	6.711	0.3806	5.671
Acidic variants	4	20.69	0	0

Aggregate reproduced to 0.8 %, which is comfortably inside the precision of the SEC method. The XMuLV log reduction factor reproduced to 5.7 %, a standard deviation of 0.38 $\log_{10}$ on a centre-point mean of 6.71 $\log_{10}$. That figure is the largest source of imprecision in the study, and it is consistent with the infectivity assay accounting for 28 % of the variance in the reported log reduction factor (Section 3.3). Acidic variants returned an identical value on every centre-point run, giving a standard deviation of 0.00 and no usable pure-error term.

The practical consequence is that lack of fit can be tested for aggregate and for the log reduction factor but not for acidic variants. The screening centre points behave the same way (Appendix A), so the pattern belongs to the responses and not to one design.

5.2 Screening: factor effects

The screening study identified the three active factors that platform knowledge predicted, and it found no significant interaction among them for any response. Table 5.2 gives the fit statistics for the three responses, and Table 5.3 and Table 5.4 give the effect estimates for the log reduction factor and for aggregate.

Table 5.2: Screening model fit statistics for the three measured responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	11	0.9919	0.9799	0.8281	82.05	0.0003856	0.03335
XMuLV LRF (log10)	11	0.9824	0.956	0.8185	37.19	0.001818	0.2648
Acidic variants	11	1	1	1	3.931e+26	1.726e-53	1.165e-14

Table 5.3: Screening effect estimates for the XMuLV log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-2.077	-1.038	0.09362	-11.09	0.0003759	***
B	1.723	0.8616	0.09362	9.203	0.0007744	***
C	0.6359	0.318	0.09362	3.396	0.02737	*
BC	0.2385	0.1192	0.09362	1.274	0.2718	
AB	0.2028	0.1014	0.09362	1.083	0.3397	
AC	-0.1939	-0.09696	0.09362	-1.036	0.3589	

Inactivation pH is the largest effect on the log reduction factor. Raising pH across its characterized range lowers the log reduction factor by 2.08 $\log_{10}$ ($p < 0.001$). Hold time is the second largest and acts in the opposite direction, raising it by 1.72 $\log_{10}$ across its range ($p < 0.001$). Temperature raises it by 0.64 $\log_{10}$ ($p < 0.05$). None of the three two-factor interactions reaches significance, and the largest of them is 0.24 $\log_{10}$ with $p = 0.27$.

Table 5.4: Screening effect estimates for pool aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.4715	0.2357	0.01179	19.99	3.693e-05	***
C	0.2209	0.1105	0.01179	9.37	0.0007227	***
AC	0.03501	0.0175	0.01179	1.485	0.2118	
AB	0.02848	0.01424	0.01179	1.208	0.2936	
A	0.02209	0.01105	0.01179	0.937	0.4018	
BC	0.01069	0.005345	0.01179	0.4534	0.6738	

Aggregate is governed by hold time and temperature, and pH has no detectable effect on it within the characterized range. Hold time raises pool aggregate by 0.47 percentage points across its range ($p < 0.001$) and temperature raises it by 0.22 points ($p < 0.001$). The pH effect is 0.022 points and does not reach significance ($p = 0.40$). That null result carries information, because the study extended pH 0.3 units below the set-point specifically in order to detect the onset of acid-induced aggregation. Over the range studied the onset was not reached, and pH is retained in the knowledge space for aggregate as evidence that the attribute is robust to it down to pH 3.2.

Acidic variants responded to hold time alone. The estimated effect is 0.4 percentage points across the characterized hold-time range, against an upper limit of 40 %, and no other term is distinguishable from zero. Because this response carries no replicate variation (Section 5.1), its fit statistics in Table 5.2 are not informative about the quality of the fit and the full effect table is given in Appendix C for completeness.

The screening study is used for identification and for nothing else. It estimates all main effects and all two-factor interactions in 11 runs, leaves 4 residual degrees of freedom, and contains no term that could detect curvature. Effect magnitudes are refined, and curvature is tested, by the response-surface models below.

5.3 Response-surface models

The response-surface models are adequate for aggregate and for the XMuLV log reduction factor, and they are the models used for the operating region, the proven acceptable ranges and the capability assessment. Table 5.5 gives the fit statistics.

Table 5.5: Response-surface model fit statistics.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	18	0.9966	0.9928	0.9607	261.4	6.566e-09	0.01695
XMuLV LRF (log10)	18	0.8961	0.7792	0.5787	7.666	0.004331	0.3665
Acidic variants	18	1	1	1	5.994e+26	2.432e-106	8.611e-15

Aggregate is modelled closely, with an R^2 of 0.997, an adjusted R^2 of 0.993 and a predicted R^2 of 0.961. The log reduction factor is modelled less closely, with an R^2 of 0.896 and a predicted R^2 of 0.579. The gap between the two responses is analytical in origin. The residual standard error of the log reduction factor model is 0.37 $\log_{10}$, and the centre-point standard deviation reported in Section 5.1 is 0.38 $\log_{10}$. The model is therefore fitting to the limit of what the infectivity assay can resolve, and a higher R^2 would not be attainable with this method.

Lack of fit is not significant for either response. Table 5.6 and Table 5.7 partition the residual sum of squares into lack of fit and pure error. For the log reduction factor the lack-of-fit F is 0.88 with p = 0.58, and for aggregate it is 1.00 with p = 0.53. Neither model departs from the data by more than the reproducibility of the scale-down system itself.

Table 5.6: Analysis of variance for the XMuLV log reduction factor model, with the residual partitioned into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.269	9	1.03	7.666	0.004331
Residual	1.075	8	0.1343		
Lack of fit	0.6402	5	0.128	0.884	0.5796
Pure error	0.4345	3	0.1448		

Table 5.7: Analysis of variance for the pool aggregate model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.6758	9	0.07509	261.4	6.566e-09
Residual	0.002298	8	0.0002872		
Lack of fit	0.001437	5	0.0002874	1.001	0.5346
Pure error	0.000861	3	0.000287		

Table 5.8 gives the coefficients of the log reduction factor model. The three main effects keep the signs and the ordering found in screening. The quadratic term in pH is negative and falls just short of significance at the 5 % level (p = 0.050). Its sign and size suggest that the surface is not linear in pH and that the decline in clearance steepens as pH rises toward the top of the range. The full quadratic model is retained for prediction, because dropping a marginal term would remove that curvature from the operating region. No other quadratic term and no interaction approaches significance.

Table 5.8: Full quadratic coefficients of the XMuLV log reduction factor model, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.907	0.1442	47.9	3.991e-11	***
A	-0.6119	0.1159	-5.279	0.0007471	***
B	0.6046	0.1159	5.216	0.0008061	***
C	0.2856	0.1159	2.464	0.03908	*
AB	0.1622	0.1296	1.252	0.246	
AC	0.06911	0.1296	0.5334	0.6083	
BC	-0.1053	0.1296	-0.8125	0.44	
A^2	-0.5132	0.2227	-2.305	0.05007	
B^2	0.2018	0.2227	0.9065	0.3912	
C^2	0.1522	0.2227	0.6837	0.5135	

The same quadratic term also turns the fitted surface over near the bottom of the pH range, where the model predicts a marginal decline in clearance at the lowest pH studied. That feature is 0.06 $\log_{10}$ in size, smaller than the residual standard error of the model by a factor of about 6, and it has no mechanistic support. It is not interpreted here and the operating region in Section 6 does not rely on the model below the set-point.

The aggregate model is dominated by its two linear terms and contains no significant curvature and no significant interaction (Appendix C). Over the characterized ranges its surface is effectively a plane in hold time and temperature.

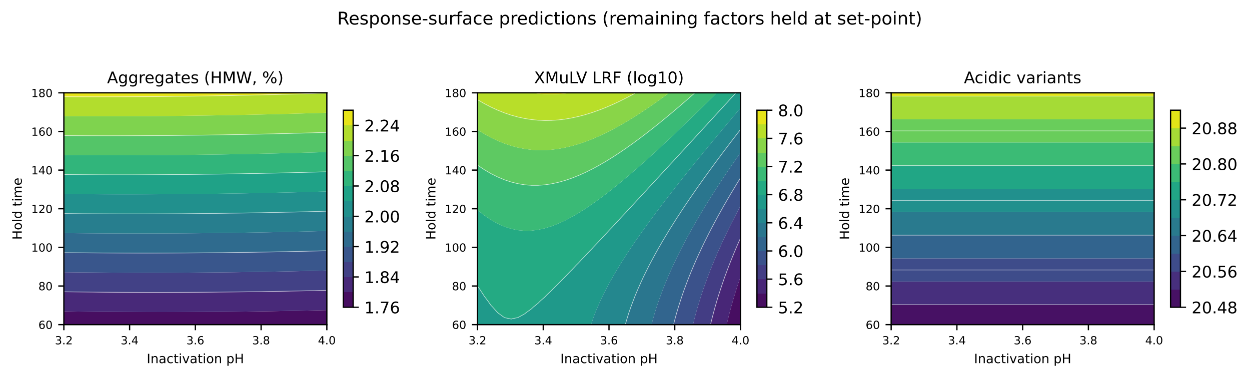

Figure 5.1: Fitted response surfaces over inactivation pH and hold time, with temperature held at the centre of its characterized range.

Figure 5.1 shows the fitted surfaces over pH and hold time. The clearance surface falls steadily from the low-pH, long-hold corner to the high-pH, short-hold corner, and the aggregate surface is almost independent of pH and rises only with hold time. The acidic variant panel shows the same hold-time dependence with no pH structure at all.

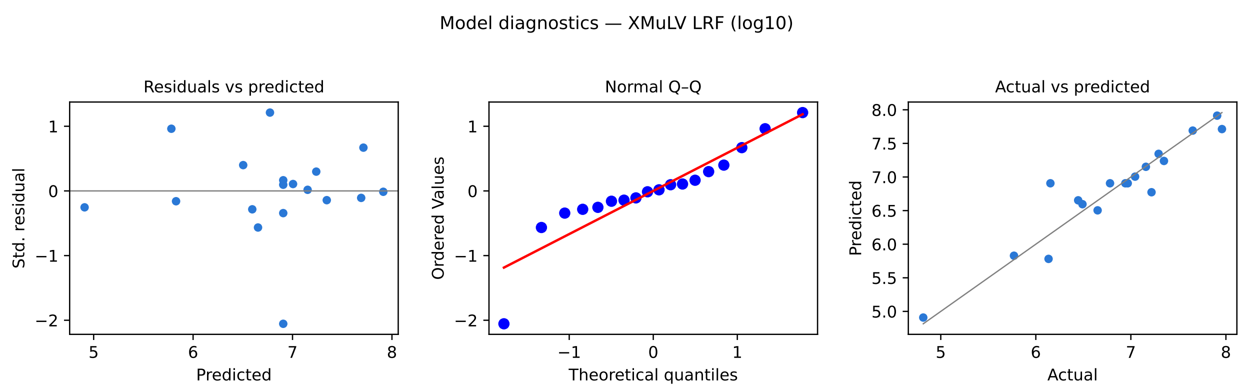

Figure 5.2: Model diagnostics for the XMuLV log reduction factor response-surface model.

Figure 5.2 gives the diagnostics for the clearance model, and one of them deserves comment before the model is used. The residuals show no structure against the predicted values, and the actual values lie about the identity line without a systematic offset at either end. The normal quantile plot, however, shows a mild departure from normality, and a Shapiro-Wilk test on the residuals returns p = 0.023. With 18 runs and a response whose reproducibility is limited by the assay, that result is not a reason to reject the model. It is a reason to treat the p-values of the marginal terms, and the pH quadratic in particular, as approximate. The largest standardized residual is 2.1, so no single run is influential enough to drive the fit. The corresponding figure for the aggregate model has a largest standardized residual of 1.6.

5.4 Mechanistic interpretation

The surfaces behave as acid denaturation kinetics predict, and that agreement is the main evidence that the models describe chemistry and not noise. Inactivation follows kinetics in which the rate constant depends on proton concentration and on temperature, and the log reduction achieved is that rate constant multiplied by the hold time. Clearance should therefore be linear in time, should fall as pH rises, and should rise with temperature. All three hold in the fitted models, and the response-surface analysis finds no significant quadratic term in hold time, which is what a first-order process requires.

The pH dependence is strongly asymmetric about the set-point, and that asymmetry is the most important single result of the study. Between the lower edge of the characterized range and the set-point the predicted log reduction factor changes by 0.03 $\log_{10}$. Between the set-point and the upper edge it falls by 1.33 $\log_{10}$. Almost all of the sensitivity of the step to pH lies above the set-point. This is consistent with acid inactivation being effectively complete within the hold once the proton concentration is high enough, and with the rate falling below what the hold can deliver as pH approaches 4.0.

Temperature acts on the rate constant in the ordinary way and its effect is the smallest of the three, because the characterized range is only 10 °C wide. Hold time acts linearly, which is why extending a hold is an effective remedy for a pH that has drifted upward within the range, and why Section 6 treats the two parameters together.

Aggregate rises with hold time and with temperature and is insensitive to pH over the range studied. Acid-induced aggregation of an IgG1 requires the molecule to be destabilized enough to expose hydrophobic surface, and the characterized range stays above the pH at which A-Mab is destabilized. What remains is a slow thermally driven association during the hold, which is why time and temperature are active and pH is not. The acidic variant response follows the same hold-time dependence, which is consistent with weak acid catalysis of asparagine deamidation over a period of hours.

The two mechanisms pull in opposite directions on hold time, and this is why the step needs a multivariate region rather than three independent ranges. A longer hold delivers more clearance and more aggregate. A hold at the top of the pH range meets the required contribution when it is long and warm, and fails it when it is short and cold: the model predicts 7.15 $\log_{10}$ at the long, warm corner of that pH and 4.91 $\log_{10}$ at the short, cold corner. No single acceptable range for pH can express that, which is what Section 6 addresses.

6 Design space

The operating region of the step is the set of combinations of inactivation pH, hold time and temperature at which the response-surface model predicts a step log reduction factor at or above 4.93 $\log_{10}$ and pool aggregate at or below 5 %. Within the characterized ranges the aggregate constraint is never binding, because the highest aggregate the model predicts anywhere in the knowledge space is 2.37 %. The region is therefore defined by the clearance constraint alone, and its principal plane is pH against hold time (Figure 5.1).

The normal operating ranges sit inside the region and the region sits inside the characterized ranges, and at the set-point the model predicts 6.87 $\log_{10}$, which is 1.94 $\log_{10}$ above the required contribution. The worst-case corner of the normal operating ranges is the highest pH, the shortest hold and the lowest temperature within them, namely pH 3.6, 60 minutes and 19 °C. At that corner the prediction is 6.43 $\log_{10}$, still 1.50 $\log_{10}$ above the requirement.

The corresponding corner of the characterized ranges is the one place in the knowledge space where the step does not deliver its contribution. At pH 4 with a 60 minute hold at 15 °C the model predicts 4.91 $\log_{10}$, which is 0.02 $\log_{10}$ below the floor. On a grid across the three characterized ranges, predictions below the required contribution occupy 0.00 % of the knowledge space and occur only at pH 4, at holds no longer than 60 minutes and at temperatures no higher than 15.5 °C. The shortfall at that corner is smaller than the residual standard error of the model, so the corner is not shown to fail so much as shown to provide no assurance of passing, and it lies outside the operating region for that reason.

Three statements bound the region, and each is a limit on what the assessor should read into it.

The region is defined by a model fitted over the characterized ranges in Table 4.1 and does not extend beyond them. Nothing in this report supports operation below pH 3.2. That lower edge is a stability boundary taken from platform data (Section 2.1) and was not demonstrated in this study, because no run approached it.

The model predicts mean responses. A condition at the edge of the region carries roughly even odds of meeting the limit on any individual batch if variability is symmetric about the mean, so the region is not the range within which the process is run. That range is the normal operating range, and the margin between the two is quantified in Section 8.

The region rests on a scale-down model qualified under SOP-1001 and is not confirmed at manufacturing scale at its edges. Confirmation at scale is limited to operation at the set-point, which is addressed by the comparability strategy in PTP-001.

Operation anywhere within the region is not a change in the regulatory sense (International Council for Harmonisation 2009). Planned movement within it still requires a prospective assessment under the pharmaceutical quality system before the move is made (International Council for Harmonisation 2008). Operation outside the region is a change that requires prior approval, and for this step it additionally requires re-assessment of the clearance claim under ICH Q5A(R2) (International Council for Harmonisation 2023a).

7 Proven acceptable ranges

The acceptance criterion applied to each attribute is stated before the ranges, because it differs between them. For aggregate and for acidic variants the criterion is the drug substance specification given in Table 2.1. For enveloped-virus clearance it is not, because that specification is cumulative across the process. The criterion applied to this step is the required step contribution, obtained by subtracting the clearance credited to anion exchange and virus filtration from the cumulative requirement of 16.7 $\log_{10}$. That leaves 4.93 $\log_{10}$ for this step, and every proven acceptable range reported for the log reduction factor is computed against that floor.

Table 7.1 gives, for each governed attribute and each multivariate parameter, the characterized range and two proven acceptable ranges. The first holds the other parameters fixed. The second varies them within their normal operating ranges by Monte-Carlo of the fitted model and requires the 95 % predictive interval of the attribute to remain within acceptance, so it is a robustness criterion and is the narrower of the two by construction.

One convention is stated here because it matters at this step. The fixed-parameter analysis holds the other factors at the coded origin of the design, which is the midpoint of each characterized range, and the same convention applies to the left-hand panels of Figure 7.1 and Figure 7.2 and to Figure 5.1. At this step the midpoint and the process set-point differ slightly for all three multivariate parameters (Table 4.1). The difference changes no range in Table 7.1, because at the process set-point values the lowest clearance predicted anywhere along the pH range is 5.54 $\log_{10}$, which is still above the floor of 4.93 $\log_{10}$.

Table 7.1: Proven acceptable ranges for each governed quality attribute and each multivariate parameter, at set-point and with the other parameters propagated across their normal operating ranges.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Inactivation pH	3.2–4	pH	3.2–4	3.2–4
Aggregates (HMW, %)	Hold time	60–180	min	60–180	60–180
Aggregates (HMW, %)	Temperature	15–25	°C	15–25	15–25
XMuLV LRF (log10)	Inactivation pH	3.2–4	pH	3.2–4	3.2–3.96
XMuLV LRF (log10)	Hold time	60–180	min	60–180	60–180

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
XMuLV LRF (log10)	Temperature	15–25	°C	15–25	15–25
Acidic variants	Inactivation pH	3.2–4	pH	3.2–4	3.2–4
Acidic variants	Hold time	60–180	min	60–180	60–180
Acidic variants	Temperature	15–25	°C	15–25	15–25

Eight of the nine entries span the whole characterized range under both analyses. For aggregate and for acidic variants no parameter is restricted at all, which follows directly from Section 5: both attributes stay far below their limits everywhere in the knowledge space, and the highest aggregate predicted anywhere is 2.37 % against a limit of 5 %.

One entry is narrower than its characterized range. The proven acceptable range for inactivation pH against the log reduction factor is the full characterized range at set-point, but 3.2–3.96 when the other two parameters vary within their normal operating ranges. The upper end falls 0.04 pH units short of the top of the characterized range. The binding attribute is enveloped-virus clearance and the mechanism is the one described in Section 5.4, which is that the clearance surface steepens as pH rises while the predictive interval widens with it.

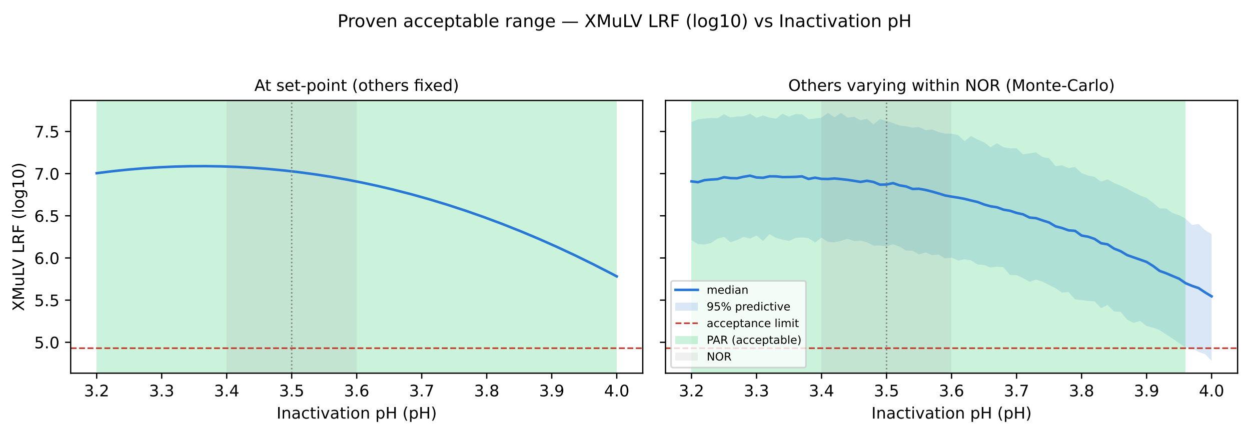

Figure 7.1: Proven acceptable range for the XMuLV log reduction factor against inactivation pH, at set-point (left) and with the other parameters propagated across their normal operating ranges (right). The acceptable region is shaded green and the normal operating range is shaded grey.

Figure 7.1 shows the analysis for pH, the governing parameter for clearance. The left panel is the fixed-parameter curve and the right panel the Monte-Carlo predictive band, with the acceptance floor drawn, the acceptable region shaded and the normal operating range marked. The set-point sits well inside the acceptable region in both panels, and the right panel shows the lower edge of the predictive band crossing the floor before the curve itself does.

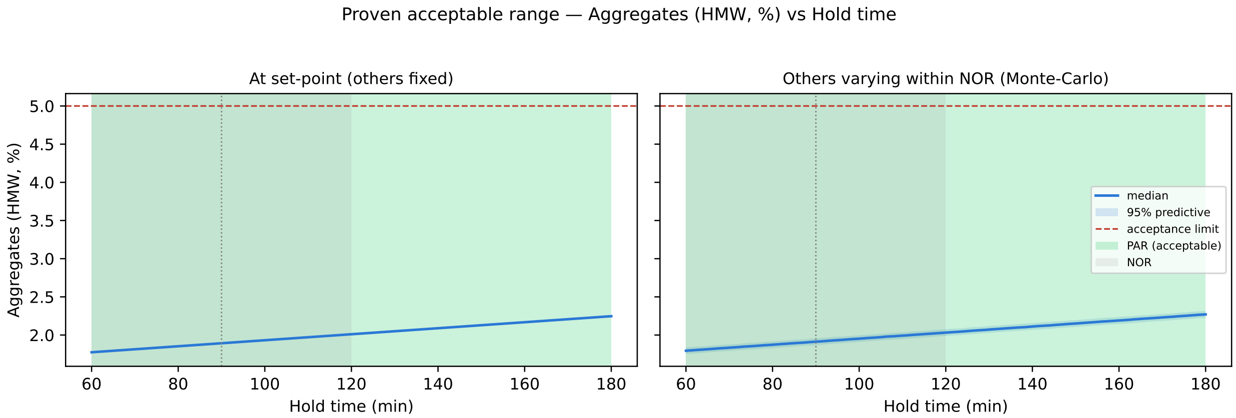

Figure 7.2: Proven acceptable range for pool aggregate against hold time, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

Figure 7.2 shows the same analysis for aggregate against hold time, the parameter that governs it. The whole characterized hold-time range is acceptable under both analyses, and even at the longest hold combined with the highest temperature the model predicts 2.37 % against a limit of 5 %.

These ranges are univariate statements and the design space is not, and the two must not be read as alternatives. Where a parameter's proven acceptable range covers its whole characterized range, the binding constraint on operation is the multivariate corner identified in Section 6 and not the univariate range. The one place where the two agree is the upper end of the pH range, where the robustness analysis and the multivariate worst-case corner point at the same edge for the same reason. Both analyses are computed from the fitted response-surface model over the characterized ranges and inherit its bounds, and neither extends outside those ranges.

8 Process capability and robustness

Capability was assessed for the three attributes this step influences, by simulating 2,000 commercial batches with every process parameter drawn within its normal operating range. Table 8.1 gives the result at drug substance.

Table 8.1: Commercial-scale capability for the attributes influenced by the low-pH viral inactivation step.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53

Enveloped-virus clearance is the tightest of the three by a wide margin. Its one-sided Cpk is 1.53 against the cumulative requirement of 16.7 $\log_{10}$, with a simulated mean of 19.51 $\log_{10}$ and a standard deviation of 0.61. The margin between the mean and the requirement is 2.81 $\log_{10}$, or 4.6 standard deviations. Aggregate and acidic variants sit far from their limits at drug substance, with Cpk values of 16.1 and 4.3 respectively, but those two figures describe the whole process and not this step alone, and the aggregate figure in particular is set by cation exchange, which is the polishing step that removes aggregate (PCR-007).

The clearance figure is likewise cumulative, and this step supplies part of it. Table 8.2 gives the modular ledger for the two model viruses.

Table 8.2: Modular viral clearance by step for XMuLV and MVM, in log10 reduction.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.9	10

This step is credited with 7.10 $\log_{10}$ of XMuLV clearance, anion exchange with 5.95 (PCR-008) and virus filtration with 5.82 (PCR-009), a cumulative 18.87 $\log_{10}$ against the requirement of 16.7. The low-pH hold supplies the largest single contribution to enveloped-virus clearance and none of the parvovirus clearance. The three mechanisms are orthogonal, which is what allows the contributions to be added under ICH Q5A(R2) (International Council for Harmonisation 2023a): chemical inactivation here, charge-based adsorption at anion exchange and size-based retention at virus filtration.

One reconciliation is needed between Table 8.2 and Section 5. The credited contribution of 7.10 $\log_{10}$ is the nominal value from the process model at set-point conditions, whereas the response-surface model of this study predicts 6.87 $\log_{10}$ at the same conditions. The two differ by 0.23 $\log_{10}$, which is smaller than the residual standard error of the response-surface model, so the credited value and the characterization data agree within the precision of the study.

Capability is conditioned on operation within the normal operating ranges and on the scale-down model. It carries no allowance for a parameter excursion outside the normal operating range, and any such excursion is assessed against the operating region in Section 6 rather than against Table 8.1.

9 Parameter classification

Every parameter of the step was classified under SOP-4001 from its demonstrated effect on quality attributes and from the risk that it leaves the operating region in routine manufacture. Table 4.1 gives the outcome and the reasoning follows.

- Inactivation pH is a critical process parameter. It has the largest effect on enveloped-virus clearance (Table 5.3), its normal operating range is only 0.2 pH units wide, and Section 7 shows that robustness to co-variation of the other parameters is lost before the top of the characterized range is reached. Unlike hold time, a pH excursion cannot be corrected once the hold has begun. It is the only parameter in the A-Mab drug substance process classified as a critical process parameter.
- Hold time is a well-controlled critical process parameter. It has the second largest effect on clearance and it governs both aggregate and acidic variants (Section 5.2), so it is linked to critical quality attributes and cannot be classified below this level. It is set by a timer, recorded automatically by the process control system, and a hold that has run short can be extended before neutralization. Its proven acceptable range covers the whole characterized range under both analyses in Section 7.
- Temperature is a well-controlled critical process parameter. Its effect on clearance is real but the smallest of the three, and it also raises aggregate. The hold is thermostatted and the range that supports the claim is 10 °C wide against a normal operating range of 4 °C, so the risk of leaving the operating region through temperature alone is low.
- A-Mab concentration is a general process parameter. No mechanism links it to acid-mediated inactivation (Section 4.4), and its normal operating range is its entire characterized range, so it places no constraint on the eluate delivered by Protein A.

No parameter of this step is a key process parameter. The step neither concentrates nor separates the product, so it has no yield or throughput parameter to control for process consistency alone. Across the whole drug substance process 1 parameter is classified as a critical process parameter, 20 as well-controlled critical process parameters, 10 as key process parameters and 6 as general process parameters, and the single critical process parameter is the pH of this step.

The absence of significant interactions is retained in the knowledge space and not discarded. No two-factor interaction reached significance for any response (Section 5.2), so within the ranges studied the three multivariate parameters can be controlled independently, and each proven acceptable range in Section 7 can be read without reference to the others except at the corner identified in Section 6.

10 Contribution to the control strategy

The step contributes a small number of controls, and they are all parameter controls rather than tests on the pool. Inactivation pH is controlled to 3.4–3.6 and is verified by an independent measurement before the hold clock starts. Hold time is controlled to 60–120 minutes and is recorded by the process control system. Temperature is controlled to 19–23 °C for the duration of the hold (SOP-2009). A-Mab concentration is monitored across 5–35 g/L and is not restricted.

The pH before the hold and the hold duration are in-process controls with acceptance at the normal operating range. A result outside the normal operating range but inside the operating region of Section 6 is dispositioned against that region and does not by itself invalidate the clearance claim, whereas a result outside the region does. This is the practical value of having characterized ranges that are between 2.0 and 4.0 times as wide as the operating ranges.

Two life-cycle controls follow from the nature of the claim. Because the clearance rests on a spiked small-scale model, any change to the acid or buffer system, to the hold configuration or to the neutralization target requires re-assessment against ICH Q5A(R2) before the credit can be retained (International Council for Harmonisation 2023a). Because the claim is cumulative, a change at anion exchange or virus filtration that reduces their credited contributions raises the floor required of this step, and the floor in Section 7 would have to be recomputed.

Release testing does not measure viral clearance and no release test can confirm it on a batch, so the claim is a process claim. That is why the parameter control described above is the control, and why the pH of this step carries the only critical process parameter classification in the process.

The step does not control on its own any attribute it influences. Enveloped-virus clearance is a cumulative claim across three orthogonal steps and this report supplies one of them, with the others in PCR-008 and PCR-009. Aggregate is reduced downstream at cation exchange (PCR-007). Acidic variants are formed at the bioreactor (PCR-003) and are of very low criticality. The consolidated position across all steps is given in PCMR-001.

11 Discussion

The step is well understood over the ranges studied, and the understanding is narrower than the volume of data might suggest. Three parameters govern a single clearance response, no interaction among them was detected, and the surfaces match the mechanism that platform experience predicted. What the study adds to that platform position is quantitative: the location of the upper pH edge of failure, the size of the margin at the set-point, and the demonstration that the product-quality cost of a long hold is small.

Confidence in the scale-down model is high for the chemistry and limited for the equipment, because the model reproduces titration, hold and neutralization with the same reagents and the same pH trajectory, and the centre-point reproducibility in Section 5.1 shows it to be as repeatable as the assay allows (Table 5.1). It says nothing about mixing homogeneity in the commercial hold vessel, and that gap is closed by equipment qualification at the receiving site under PTP-001, not by this report.

Two deviations occurred and both were retained (Section 13). Neither is a protocol excursion in the sense of a design point executed at the wrong setting. One was a record discrepancy of 4 minutes on a hold-time entry, worth 0.040 $\log_{10}$ of clearance on the fitted model. The other raised the working quantitation limit of the infectivity assay for a subset of samples, which makes the affected log reduction factors lower bounds and therefore makes the fitted model conservative. Neither changed a classification or the operating region.

Four limitations bound what has been shown, and they are stated together because they compound. The clearance claim rests on a spiked small-scale model and will never be confirmed on a manufacturing batch. The response-surface model predicts mean responses, so a condition at the edge of the operating region provides no assurance for an individual batch. The operating region is not confirmed at scale at its edges. The lower edge of the pH range is a platform boundary rather than a demonstrated one, so the two-sided character of the pH constraint is demonstrated on one side and inherited on the other.

A fifth limitation is analytical and it sets the resolution of everything above. The infectivity assay contributes 28 % of the variance in the reported log reduction factor, and the residual standard error of the fitted model, 0.37 $\log_{10}$, is close to the centre-point standard deviation of 0.38 $\log_{10}$. The step's clearance is therefore known to a few tenths of a log, which is comfortable against a set-point margin of 1.94 $\log_{10}$ and uncomfortable against the 0.02 $\log_{10}$ shortfall at the worst characterized corner. The report treats the first as a real margin and the second as an absence of assurance rather than a demonstrated failure.

The classification of pH as a critical process parameter was necessary, and the study is what makes it necessary. Hold time and temperature have real effects on a critical quality attribute, so they are critical parameters in the sense of ICH Q8, but each is measured continuously, each can be corrected before neutralization, and each retains its full characterized range as a proven acceptable range under co-variation. pH satisfies none of those three conditions. That difference, and not the size of the effect, is what separates the two classes at this step.

12 Conclusions

The low-pH viral inactivation step is well characterized over the ranges studied and is robust within its normal operating ranges. A response-surface model in inactivation pH, hold time and temperature predicts the XMuLV log reduction factor with a residual standard error of 0.37 $\log_{10}$ and no significant lack of fit, and it defines an operating region in which the step delivers at least its required contribution of 4.93 $\log_{10}$. At the set-point the predicted contribution is 6.87 $\log_{10}$.

Proven acceptable ranges cover the whole characterized range for every parameter and attribute except inactivation pH against clearance, which is restricted to 3.2–3.96 under co-variation of the other parameters within their normal operating ranges. At commercial scale the cumulative enveloped-virus clearance has a one-sided Cpk of 1.53 against a requirement of 16.7 $\log_{10}$, which is the tightest of the three capabilities this step influences. Aggregate and acidic variants remain far from their limits throughout the knowledge space.

Inactivation pH is a critical process parameter and the only one in the drug substance process. Hold time and temperature are well-controlled critical process parameters, and A-Mab concentration is a general process parameter. Two deviations were recorded and both were retained without effect on the models, the operating region or the classifications.

The step supplies enveloped-virus clearance and nothing else. It is credited with no parvovirus clearance and no host cell protein clearance, and the cumulative viral safety of the drug substance depends equally on anion exchange (PCR-008) and virus filtration (PCR-009). The results of this report roll up into PCMR-001 and support Stage 2 operation at the receiving site within the ranges given in Table 4.1.

13 Deviations from the plan

Two deviations were recorded during execution of PCP-006. Each was investigated to root cause, impact-assessed against the fitted models and closed, and both were retained. Table 13.1 is the register.

Table 13.1: Register of deviations recorded during execution of PCP-006.

Deviation	Summary	Detected during	Disposition
DEV-006-01	Hold-time record discrepancy between DCS and manual log	batch-record reconciliation	retained
DEV-006-02	XMuLV infectivity assay cytotoxicity at low dilution	assay review	retained

Neither deviation invalidated a run, and no design point was re-executed. The two entries differ in kind: the first concerns a record and the second concerns an assay, and neither concerns the process itself.

13.1 DEV-006-01 — hold-time record discrepancy

The hold duration recorded for one run by the process control system differed from the duration entered in the manual batch log by 4 minutes. The discrepancy was found during batch-record reconciliation after execution and not during the run, so the run itself proceeded to its programmed end point.

The investigation traced the difference to a clock synchronisation offset between the process control system and the manual recording station. The offset was constant across the campaign and was corrected before any further execution, and the control system record was taken as the source of truth because it is the timed record referenced by SOP-2009 and it is the record that drives the hold.

The impact is bounded by the design of the study. The discrepancy is 3.3 % of the characterized hold-time range. On the fitted response-surface model it corresponds to 0.040 $\log_{10}$ of clearance and 0.016 percentage points of aggregate, both an order of magnitude below the residual standard error of the respective model and far below the centre-point reproducibility in Table 5.1. The run was retained with its control system duration, and re-fitting is not warranted because a change of that size cannot move a coefficient beyond its standard error.

13.2 DEV-006-02 — infectivity assay cytotoxicity at low dilution

Cytotoxicity was observed in the indicator cell monolayer at the lowest dilutions of the neutralized hold pool samples, which prevented the titre being read at those dilutions. The finding came out of analytical review of the assay plates, and it applied to the pool samples but not to the spiked load samples.

The investigation attributed the cytotoxicity to the sample matrix. The neutralized hold buffer is the most concentrated matrix presented to the assay in this study, and the effect was absent in the load samples, which are read at a higher dilution because their titre is higher. The virus preparation and the cell bank were excluded as causes by the assay controls required under AMV-3017 (back-titration, cell control and positive control).

The disposition was to report titres from a dilution of 8 and above, at which no cytotoxicity was observed. Doing so raises the working quantitation limit of the assay for the affected samples above the validated limit of 1.5 $\log_{10}$ TCID50/mL. The consequence for the study is a floor on the measurable pool titre and not a bias in it. A pool titre reported at a raised floor understates the clearance achieved, so the affected log reduction factors are lower bounds and the fitted model is conservative in that direction.

No run was excluded from the analysis and no titre was replaced by a substituted value. The cytotoxicity control is a standing part of the method and its performance is summarized in Appendix D. The finding is recorded here because it is a limitation on the resolution of the clearance measurement, and because it is one of the reasons the study reports a margin at the set-point and does not report a maximum achievable clearance.

14 Appendices

The appendices carry the full design matrices, the complete effect and coefficient tables for every response, and the analytical method summary. They are the record behind the tables in Section 5 and are not summarized again here.

Appendix A — Screening design matrix and responses

The coded screening matrix and its measured responses, in run order. Coded −1 and +1 are the edges of each characterized range and 0 is the range midpoint.

Table 14.1: Screening design matrix in coded factors, with measured responses.

Run	Type	A	B	C	Aggregates (HMW, %)	XMuLV LRF (log10)	Acidic variants
1	factorial	-1	-1	-1	1.694	7.023	20.49
2	factorial	-1	-1	1	1.903	7.483	20.49
3	factorial	-1	1	-1	2.16	8.174	20.89
4	factorial	-1	1	1	2.323	9.373	20.89
5	factorial	1	-1	-1	1.687	4.806	20.49
6	factorial	1	-1	1	1.898	5.141	20.49
7	factorial	1	1	-1	2.142	6.625	20.89
8	factorial	1	1	1	2.443	7.174	20.89
9	center	0	0	0	2.01	7.351	20.69
10	center	0	0	0	2.066	6.916	20.69
11	center	0	0	0	2.01	7.356	20.69

Table 14.2: Centre-point reproducibility of the screening design.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	3	2.028	0.0324	1.597
XMuLV LRF (log10)	3	7.208	0.2527	3.506
Acidic variants	3	20.69	0	0

Appendix B — Response-surface design matrix and responses

The coded face-centred central composite matrix and its measured responses. Axial runs sit at the face centres, so no run leaves the characterized ranges.

Table 14.3: Response-surface design matrix in coded factors, with measured responses.

Run	Type	A	B	C	Aggregates (HMW, %)	XMuLV LRF (log10)	Acidic variants
1	factorial	-1	-1	-1	1.687	6.491	20.49
2	factorial	-1	-1	1	1.882	7.348	20.49
3	factorial	-1	1	-1	2.17	7.651	20.89
4	factorial	-1	1	1	2.356	7.908	20.89
5	factorial	1	-1	-1	1.676	4.816	20.49
6	factorial	1	-1	1	1.881	5.77	20.49
7	factorial	1	1	-1	2.111	6.446	20.89
8	factorial	1	1	1	2.393	7.158	20.89
9	axial	-1	0	0	2.007	7.044	20.69
10	axial	1	0	0	2.016	6.134	20.69
11	axial	0	-1	0	1.786	6.65	20.49
12	axial	0	1	0	2.243	7.958	20.89
13	axial	0	0	-1	1.924	7.217	20.69
14	axial	0	0	1	2.129	7.292	20.69
15	center	0	0	0	2.021	6.942	20.69
16	center	0	0	0	1.983	6.153	20.69
17	center	0	0	0	2.016	6.967	20.69
18	center	0	0	0	2.003	6.781	20.69

Appendix C — Full effect and coefficient tables

Screening effect estimates for the response not tabulated in Section 5.2, followed by the full quadratic coefficients for the two responses not tabulated there.

Table 14.4: Screening effect estimates for acidic variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.4	0.2	4.12e-15	4.86e+13	1.08e-54	***
BC	-6.19e-15	-3.09e-15	4.12e-15	-0.751	0.494	
C	-5.35e-15	-2.67e-15	4.12e-15	-0.649	0.551	
AB	-4.57e-15	-2.29e-15	4.12e-15	-0.555	0.608	
AC	-2.6e-15	-1.3e-15	4.12e-15	-0.316	0.768	
A	6.69e-16	3.35e-16	4.12e-15	0.0812	0.939	

Table 14.5: Full quadratic coefficients of the pool aggregate model.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.01	0.006667	301.5	1.642e-17	***
A	-0.002438	0.005359	-0.4549	0.6613	
B	0.2362	0.005359	44.07	7.757e-11	***
C	0.1074	0.005359	20.04	4.013e-08	***
AB	-0.001327	0.005992	-0.2214	0.8303	
AC	0.01326	0.005992	2.213	0.05781	
BC	0.008374	0.005992	1.398	0.1998	
A^2	-0.002473	0.0103	-0.2402	0.8162	
B^2	0.0006272	0.0103	0.06091	0.9529	
C^2	0.01246	0.0103	1.211	0.2606	

Table 14.6: Full quadratic coefficients of the acidic variant model.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	20.7	3.39e-15	6.11e+15	5.79e-124	***
A	2.63e-15	2.72e-15	0.964	0.363	
B	0.2	2.72e-15	7.34e+13	1.32e-108	***
C	2.28e-15	2.72e-15	0.838	0.426	
AB	-1.19e-15	3.04e-15	-0.391	0.706	
AC	-2.19e-15	3.04e-15	-0.719	0.493	
BC	-3.12e-15	3.04e-15	-1.02	0.335	
A^2	-4e-15	5.23e-15	-0.764	0.467	
B^2	2.22e-15	5.23e-15	0.424	0.682	
C^2	-1.78e-15	5.23e-15	-0.34	0.743	

Appendix D — Analytical methods summary

Validated methods used for the study responses, and the reported performance of the infectivity assay that carries the clearance claim.

Table 14.7: Reported performance of the XMuLV infectivity assay (AMV-3017).

Method	Title	Precision	LOQ	LOQ unit	Accuracy (%)	Variance fraction
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5	log10 TCID50/mL	96	0.28

The remaining methods listed in Table 3.1 are the corpus-wide size-variant and charge-variant assays, and their validation records are cited there. Aggregate is the reportable result of SEC-HPLC under AMV-3011 and acidic variants the reportable result of icIEF under AMV-3013.

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